M2 Model Update

Feature Selection + Position Sizing | March 2026

What We Changed and Why

The original M2 model used all 22 features including VIX and OAS regime indicators, and made a binary yes/no decision on each M1 signal using a fixed 0.5 probability threshold. Two problems were identified:

Problem 1: Wrong features

EDA showed VIX and OAS have no independent signal (chi-square p=0.89, p=0.77). They are redundant with M1's own macro indicators. Keeping them added noise.

Problem 2: Binary gate costs returns

When M2 rejected a trade, the portfolio earned 0. But M1 wins 72% of the time, so most rejected months had positive returns that were simply missed. This created a negative Information Ratio even when selected trades looked good.

Fix 1 - Feature Selection (M2 v2)

Removed the 10 VIX and OAS features, keeping only the 12 core M1 context features: M1 direction, composite score, the 6 macro z-scores, and M1's trailing track record. This reduced noise without losing meaningful signal.

We also swept thresholds from 0.10 to 0.90 in 0.01 steps to find the optimal cutoff. The binary gate at threshold 0.51 approved 28 of 78 trades and achieved Sharpe 2.34 on those selected trades -- but the IR remained negative (-0.607) because 50 months sat in cash earning nothing.

Fix 2 - Position Sizing Instead of Binary Gate (M2 v3)

Rather than yes/no, M2's probability now directly scales how much of each trade is taken. The sizes are normalised so the average exposure equals M1:

```
position_size = m2_prob / mean(m2_prob)

m2_prob = 0.73  ->  size = 1.16x  (above average confidence, trade more)
m2_prob = 0.55  ->  size = 0.88x  (below average confidence, trade less)
m2_prob = 0.35  ->  size = 0.56x  (low confidence, trade much less)
```

Every trade still participates. High confidence months are amplified (up to 1.72x). Low confidence months are reduced (down to 0.35x). No month earns zero. The opportunity cost of the binary gate is eliminated entirely.

Results

Setup	Trades	Ann Return	Sharpe	Info Ratio
Setup	N	Ann Return	Sharpe	Info Ratio
M1 baseline (no filter)	**78**	**9.53%**	**1.310**	**+0.000**
M2 v1 - Binary Gate (22 features incl. VIX/OAS)				
t=0.30	69	9.10%	1.283	-0.236
t=0.40	56	6.98%	1.056	-0.737
t=0.50	46	5.42%	0.891	-0.937
t=0.60	38	6.19%	1.146	-0.634
t=0.70	25	5.22%	1.161	-0.705
t=0.80	10	1.89%	0.924	-1.062
M2 v2 - Binary Gate (12 core features, no VIX/OAS)				
t=0.30	73	9.34%	1.371	-0.072
t=0.40	58	6.13%	1.003	-0.781
t=0.50	31	5.14%	0.990	-0.801
t=0.51	28	6.01%	1.233	-0.607
t=0.60	16	4.84%	1.196	-0.729
t=0.70	5	1.89%	0.660	-1.109
t=0.80	2	1.05%	0.453	-1.210
M2 v3 - Position Sizing (12 core features)				
Normalised sizing (avg=1.0x)	**78**	**10.84%**	**1.345**	**+0.457**

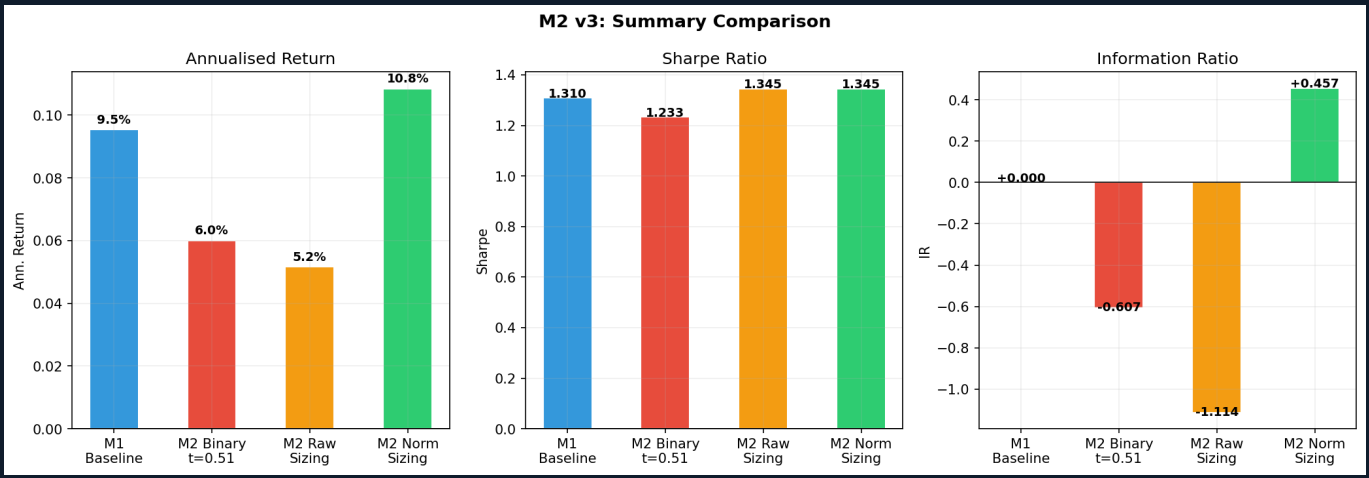

Figure 1: Ann Return, Sharpe, and IR across all four setups. Green = M2 v3 (normalised sizing).

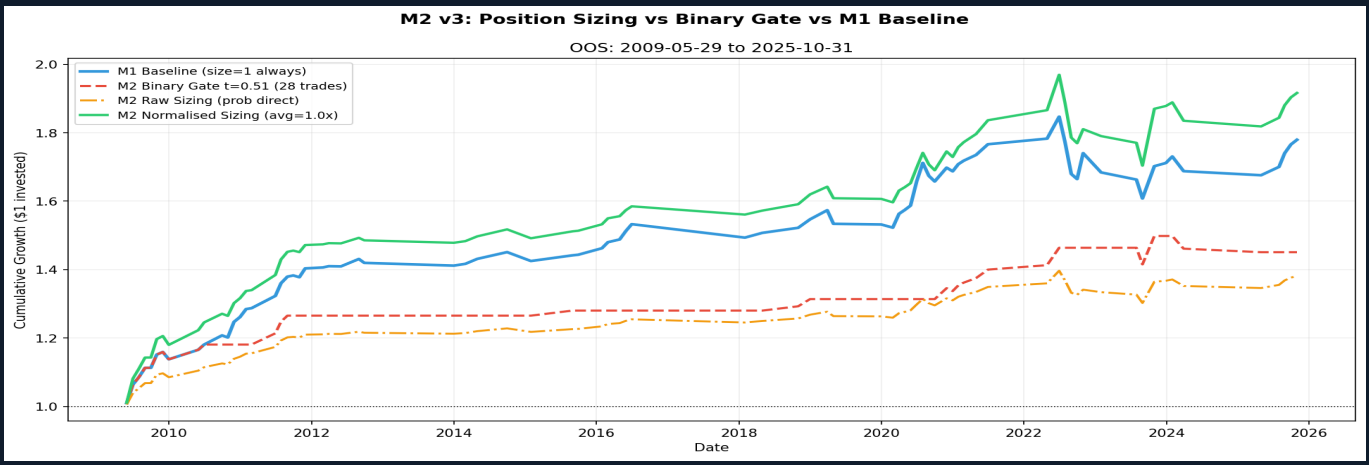

Figure 2: Cumulative return 2009-2025. M2 normalised sizing (green) consistently leads M1 baseline (blue).

Key Takeaway

Position sizing is the first M2 configuration to beat M1 on all three metrics simultaneously: higher annual return (+1.3%), higher Sharpe (1.345 vs 1.310), and a positive Information Ratio (+0.457 vs 0.000). The next step is adding features that are orthogonal to M1 to give M2 genuinely new information.